Memory Managmennt Keypoints:

MM handles how memory is organized and allocated, while AT handles how addresses are mapped through that organization.Translating a logical address into a physical one using the page table.

- [ ] 1. Paging in Memory Management (Allocation Stage)

In the Memory Management part , paging is used to allocate memory to processes.

The OS divides physical memory into fixed-size blocks called frames.

Each process's logical memory is also divided into equal-size pages.

The OS assigns available frames to those pages when the process loads into memory.

So here, paging = how memory is distributed (allocated).

📘 Example:
If Process A needs 4 pages, and we have 10 free frames —
the OS assigns frames [0, 1, 2, 3] to those 4 pages.
That's memory management using paging.

- 2. Paging in Address Translation (Access Stage)

Once memory is allocated, the CPU needs to access data in memory during execution.
However, the CPU generates logical (virtual) addresses — not physical ones.

The address translation mechanism uses the same paging concept:

It splits the logical address into page number + offset.

The page number is used to look up the frame number in the page table.

The physical address is then computed as:
Physical Address = (Frame Number * Page Size) + Offset

So here, paging = how memory is accessed safely.

- [ ] Understanding Frame Calculation in Real Systems

In an actual operating system, the number of physical frames is determined by the size of main memory and the page size used by the system.

Formula:
Number of Frames = (Physical Memory Size) / (Page Size)

Example:
If a system has 4 GB of RAM and a page size of 4 KB:
Number of Frames = (4 × 1024 × 1024 KB) / 4 KB = 1,048,576 frames

In real systems (Linux):
To find these values practically:

Command to check page size:
getconf PAGE_SIZE

Command to check total memory:
grep MemTotal /proc/meminfo

Divide the total memory by the page size to find the approximate number of frames available in physical memory.